

# Advancing Self-Improving Agents through Long-Horizon *Reflective Tasks*

Learning to turn more attempts into better solutions.

Train reflection where feedback is clear. Carry improvement beyond the training ground.

IN DOMAIN

81.8

MLE-bench Lite

Medal Average

IN DOMAIN

70.7

Frontier-CS

Agent Track · 188 tasks

CROSS-DOMAIN

84.0

BrowseComp

Accuracy

CROSS-DOMAIN

52.6

HLE

Text-only accuracy

Qwen3.8-27B backbone. Scores from the accompanying manuscript; benchmark-specific protocols apply.

## ABSTRACT

We present **AREX 2.0**, an effort to advance the self-improving capability of LLM agents, defined as the ability to iteratively refine a solution at test time. This ability rests on two capabilities: *reflection*, which produces a better solution than the current one, and *long-horizon execution*, which sustains effective iteration. We hypothesize that both are domain-agnostic and can be learned from scenarios well suited for supervision. We therefore synthesize long-horizon improvement trajectories from machine learning and algorithmic programming tasks, which offer verifiable feedback and reward sustained iteration. Trained on this data, our Qwen3.8-27B-based agent achieves strong results on MLE-bench Lite (81.8) and Frontier-CS (70.7), transfers to deep research with 84.0 on BrowseComp and 52.6 on HLE, and improves more steeply as iterations increase. These results show that long-horizon reflective data is an effective route to self-improving agents.

# Better at improving. Not just better on the first try.

We study an agent's ability to iteratively improve a solution **within a single task, at test time**. The agent must decide what to change, use the feedback it receives, and turn additional rounds into better outcomes.

A retry loop can live in a scaffold. Our goal is to internalize the capability that makes the loop useful: judging a solution's shortcomings and deciding how to revise it.

That requires both **reflection**, the expected gain of an individual revision, and **long-horizon execution**, the ability to sustain productive revisions over time.

$$\mathbb{E}[s_T] \approx s_0 (1 + r)^T$$

A heuristic for expected improvement, not a law.

*r*

## Reflection

Expected relative gain from one round.

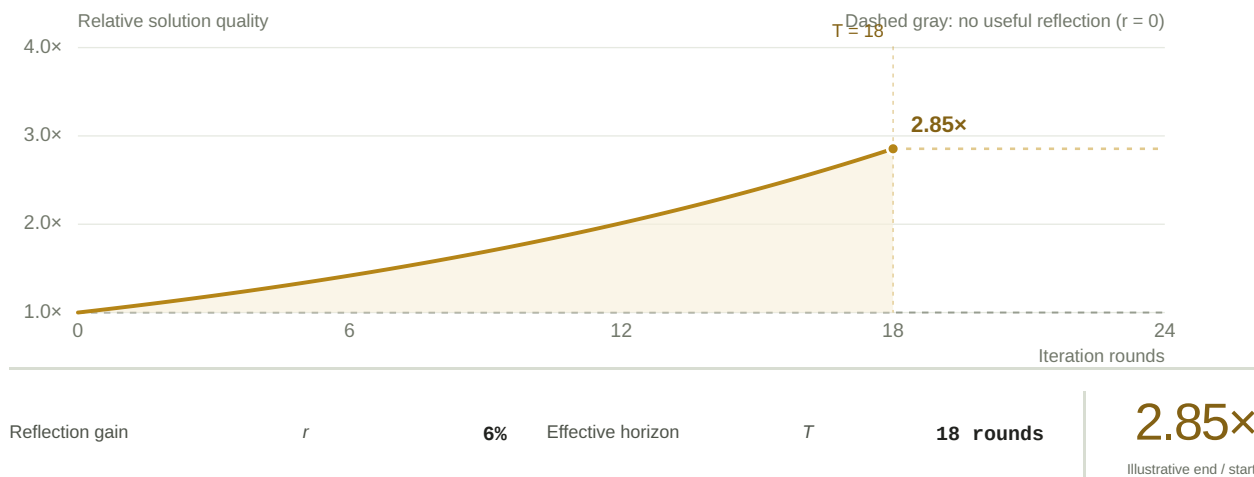
*T*

## Long-horizon execution

How many rounds remain effective.

### Two capabilities, one compounding loop.

ILLUSTRATION · NOT MEASURED RESULTS



For positive initial scores. The curve shows an idealized expectation, not a sampled trajectory or a prediction of benchmark performance. Individual rounds can regress, and real scores have ceilings.

**Both capabilities matter.** When *r* is near zero, more rounds go nowhere. When *T* is small, even good reflection has little time to compound. Here, self-improvement does not mean accumulating memory or changing weights across test-time tasks.

# Teach the journey, not only the destination.

Long-horizon reflection needs to be learned. Yet a corpus of successful answers does not, by itself, show how an agent gets from a weak solution to a strong one through measurement, setbacks, and revision.

THE COMMON UNIT OF SUPERVISION

## A successful attempt

Task → solution → pass.

The destination is kept; the improvement process can disappear.

THE UNIT WE WANT TO PRESERVE

## A complete improvement trajectory

Try → measure → reflect → revise → continue.

The agent learns how to recover, not just how to succeed.

Our hypothesis is that **long-horizon reflection is a domain-agnostic meta-skill**. The right training ground is therefore not necessarily the one closest to the target task, but the one that provides the clearest supervision for improvement.



SOURCE / GITHUB

## Machine learning

|                 |                                                                            |
|-----------------|----------------------------------------------------------------------------|
| <b>Feedback</b> | Validation metrics make the effect of a revision measurable.               |
| <b>Headroom</b> | Baselines can improve through experiments, tuning, and stronger pipelines. |
| <b>Material</b> | Repositories supply code, data workflows, and evaluation logic.            |



SOURCE / ONLINE JUDGES

## Algorithmic programming

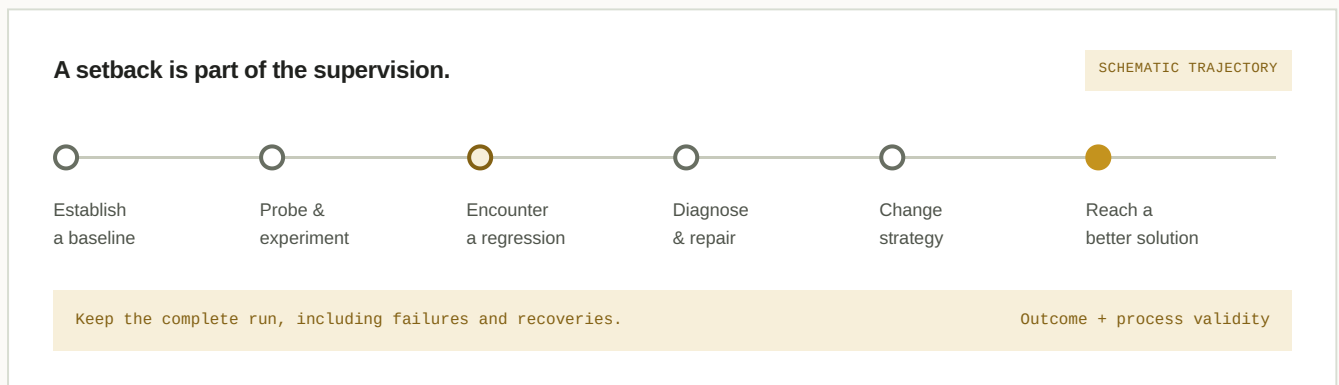
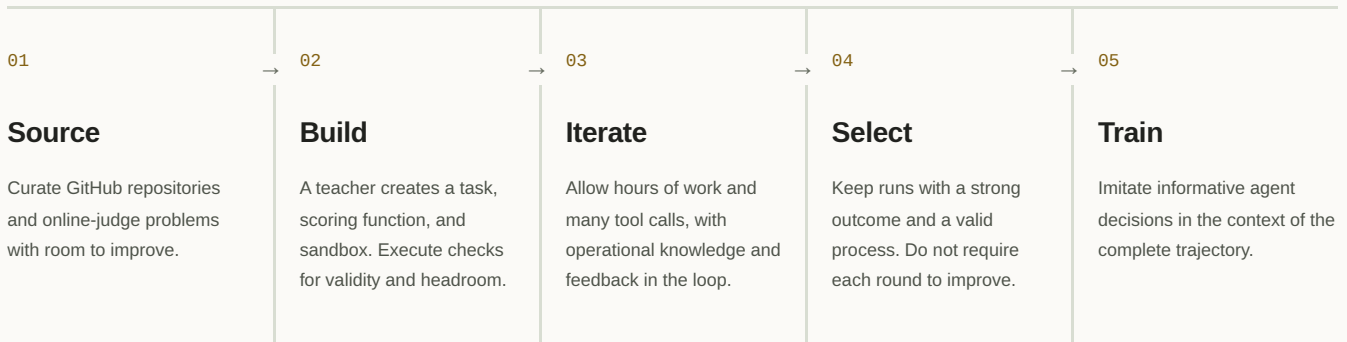
|                 |                                                                           |
|-----------------|---------------------------------------------------------------------------|
| <b>Feedback</b> | Tests and graded scoring functions verify solution quality.               |
| <b>Headroom</b> | Programs can improve in correctness, efficiency, or optimization quality. |
| <b>Material</b> | Problem statements, reference solutions, and tests support synthesis.     |

THE TRANSFER HYPOTHESIS

Learn how to improve where the signal is clean.  
Retain that ability when the domain changes.

# Build environments. Keep the whole trajectory.

We turn repositories and judge problems into executable training environments, then collect long runs in which a teacher agent has the time, knowledge, and feedback to improve.



## Selection is at the run level.

A selected trajectory must deliver a strong result and respect the task's requirements. Failed experiments, reversions, and abandoned approaches remain. Filtering them out would remove the very behavior needed when a round goes wrong.

## Loss is on meaningful decisions.

Supervised targets include hypotheses, experiments, diagnoses, repairs, strategy changes, and submissions. Tool observations and retrieved material provide context; redundant polling and near-duplicate turns receive no loss.

New ML + coding trajectories + unchanged AREX deep-research data

→ AREX 2.0 · 27B

The teacher's operational knowledge helps produce trajectories worth imitating. The trained model learns the behavior through its weights; this is distinct from accumulating skills or memory across evaluation tasks.

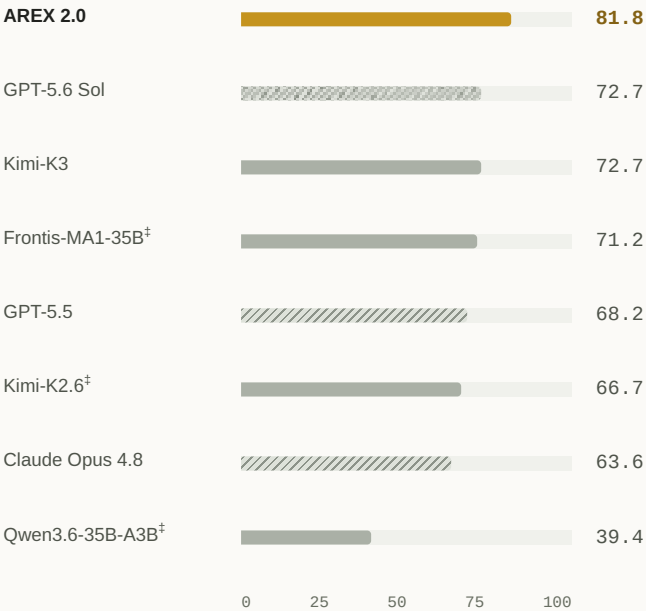
# Strong where the trajectories are built.

The first test is whether the model learns the capabilities demonstrated in the new data. AREX 2.0 performs strongly in both machine learning engineering and algorithmic programming.

AREX 2.0   Open-weight baseline   Closed-weight baseline

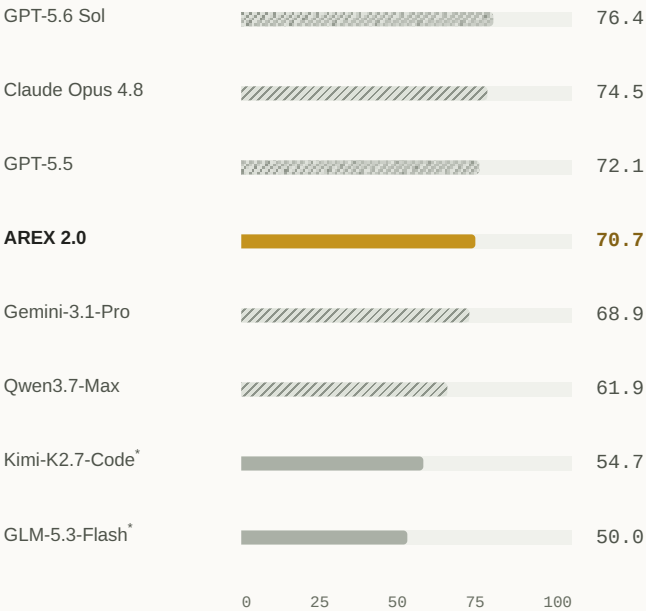
## MLE-bench Lite

Medal Average · higher is better



## Frontier-CS

188-task Agent Track · higher is better



Selected comparisons from the manuscript. All bar axes start at zero. \* Reproduced by the authors. ‡ Reproduced and reported by OpenMLE. The full comparison tables appear at the end of this overview.

### 81.8 on MLE-bench Lite

The highest Medal Average in the reported comparison set: **9.1 points above** the best listed baseline score of 72.7.

### 70.7 on Frontier-CS

Above every listed open-weight baseline and **5.7 points behind** the highest reported score of 76.4.

#### MACHINE LEARNING ENGINEERING

##### MLE-bench Lite / MLE-Lite

OpenMLE evaluation protocol.

Maximum budget: 12 hours per task.

#### ALGORITHMIC PROGRAMMING

##### Frontier-CS / Agent Track

188 tasks in the reported evaluation.

Maximum budget: 5 hours per task.

# Trained to improve here. Better at research there.

No deep-research trajectories are added by the new pipeline. Yet the resulting model also improves on search and reasoning benchmarks—a pattern consistent with the transfer hypothesis.

BrowseComp

84.0

Highest listed ≤40B score  
Accuracy

HLE

52.6

Highest listed ≤40B score  
Text-only accuracy

GAIA

92.2

Below Agents-A1's 96.0  
Accuracy

DeepSearchQA

93.8

Highest listed ≤40B score  
F1

## Across generations of AREX

| Model                     | Size       | BrowseComp  | HLE         | GAIA        | DeepSearchQA |
|---------------------------|------------|-------------|-------------|-------------|--------------|
| AREX-Turbo                | 4B         | 70.7        | 40.6        | 81.6        | 78.5         |
| AREX-Base                 | 122B       | 82.5        | 52.4        | 85.4        | 89.9         |
| <b>AREX 2.0</b>           | <b>27B</b> | <b>84.0</b> | <b>52.6</b> | <b>92.2</b> | <b>93.8</b>  |
| Difference vs. AREX-Turbo |            | +13.3       | +12.0       | +10.6       | +15.3        |

Relative to AREX-Turbo, the new model gains **13.3 points on BrowseComp** and **12.0 on HLE**. It also exceeds the 122B AREX-Base on all four reported research benchmarks.

The new trajectories come from machine learning and algorithmic programming, while the original deep-research corpus is carried over unchanged. Gains beyond the training domains are consistent with learning a transferable capacity to improve.

**Interpreting transfer.** The deep-research data is reused unchanged. Cross-generation comparisons also differ in backbone and parameter count, however; isolating the effect of the new trajectories requires a matched-backbone ablation. The reported pattern supports, rather than by itself proves, the transfer hypothesis.

Non-coding evaluations follow the AREX protocol: up to 300 inner turns and 1,500 overall turns. HLE values in this AREX-only comparison use the text-only subset. Full-set HLE results are marked separately in the appendix and are not directly comparable.

# More rounds should earn their keep.

A higher final score is only part of the story. The defining behavioral question is whether an agent converts a larger iteration budget into a better solution—and keeps doing so when an experiment fails.

- 
- |    |                                |                                                                                                                                                      |
|----|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | <b>Make reflection useful.</b> | Feedback reveals where the current solution falls short. Operational knowledge helps the agent turn that signal into a revision that can improve it. |
|----|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
- 
- |    |                                  |                                                                                                                                                |
|----|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 02 | <b>Keep iteration effective.</b> | Long-horizon execution preserves what earlier rounds taught, responds to regressions, and continues searching rather than abandoning the task. |
|----|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
- 
- |    |                                      |                                                                                                                                          |
|----|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 03 | <b>Measure the return on rounds.</b> | A steeper improvement curve corresponds to a larger effective $r$ ; useful gains sustained over more rounds correspond to a larger $T$ . |
|----|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
- 

The study reports stronger improvement as iterations increase, with feedback, operational knowledge, and additional rounds contributing to the behavior. This overview visualizes the reported endpoint benchmark scores; the earlier  $r/T$  curve is a mechanism illustration, not an empirical score–iteration trace.

## THE TAKEAWAY

Do not only teach agents what works.  
Teach them *how to keep improving*.

Long-horizon reflective data is an effective route toward self-improving agents. Building that data in machine learning and algorithmic programming gives models a training ground for reflection and persistence—and the reported results suggest those capabilities can travel beyond it.

**About this overview.** Based on the accompanying manuscript, *Advancing Self-Improving Agents through Long-Horizon Reflective Tasks*. All benchmark values are taken from that manuscript. Comparisons refer to its reported model set, and model size means total parameters. Mechanism diagrams are illustrative.

# Programming & machine learning

Frontier-CS on the 188-task Agent Track, and Medal Average on MLE-bench Lite. A dash indicates an unreported value.

| Model                          | Size | Frontier-CS       | MLE-Lite          |
|--------------------------------|------|-------------------|-------------------|
| CLOSED-WEIGHT MODELS           |      |                   |                   |
| GPT-5.6 Sol                    | —    | 76.4              | 72.7              |
| Claude Opus 4.8                | —    | 74.5              | 63.6              |
| GPT-5.5                        | —    | 72.1              | 68.2              |
| Gemini-3.1-Pro                 | —    | 68.9              | —                 |
| Qwen3.7-Max                    | —    | 61.9              | —                 |
| OPEN-WEIGHT MODELS             |      |                   |                   |
| Kimi-K3                        | 2.8T | —                 | 72.7              |
| DeepSeek-V4-Pro <sup>†</sup>   | 1.6T | 44.7 <sup>*</sup> | 54.5 <sup>†</sup> |
| DeepSeek-V4-Flash <sup>†</sup> | 284B | 39.1 <sup>*</sup> | 51.5 <sup>†</sup> |
| Kimi-K2.7-Code <sup>†</sup>    | 1T   | 54.7 <sup>*</sup> | —                 |
| GLM-5.3-Flash <sup>†</sup>     | 320B | 50.0 <sup>*</sup> | —                 |
| Kimi-K2.6                      | 1T   | 46.9              | 66.7 <sup>‡</sup> |
| Frontis-MA1-35B                | 35B  | —                 | 71.2 <sup>‡</sup> |
| BigBang-V1                     | 35B  | —                 | 59.1              |
| Qwen3.6-35B-A3B                | 35B  | 23.4              | 39.4 <sup>‡</sup> |
| AREX 2.0                       | 27B  | 70.7              | 81.8              |

**Evaluation notes.** \* Results reproduced by the authors. ‡ Results reproduced and reported by OpenMLE. † Markers are retained from the manuscript; its caption identifies default pi-agent evaluation for the corresponding Frontier-CS results. The same marker also appears on two MLE-Lite entries in the source table, without a separate definition for that use.

Model names, sizes, values, and provenance markers follow the supplied manuscript. Rankings are within this comparison set. The budgets used for AREX 2.0 are 5 hours per Frontier-CS task and 12 hours per MLE-Lite task.

# Deep research & agentic reasoning

BrowseComp, HLE, and GAIA report accuracy; DeepSearchQA reports F1. Model size refers to total parameters.

| Model                   | Size | BrowseComp | HLE   | GAIA | DeepSearchQA |
|-------------------------|------|------------|-------|------|--------------|
| FRONTIER MODELS         |      |            |       |      |              |
| GPT-5.6 Sol             | —    | 90.4       | 58.0* | —    | —            |
| Kimi-K3                 | 2.8T | 91.2       | 56.0* | —    | 95.0         |
| Claude Fable 5          | —    | 88.0       | 64.5* | —    | 94.2         |
| Claude Opus 4.8         | —    | 84.3       | 57.9* | —    | 93.1         |
| GPT-5.5                 | —    | 84.4       | 52.2* | 87.4 | —            |
| Gemini-3.1-Pro          | —    | 85.9       | 51.4* | 80.6 | 93.3         |
| LARGE MODELS (>40B)     |      |            |       |      |              |
| GLM-5                   | 744B | 75.9       | 50.4  | 70.0 | —            |
| Kimi-K2.6               | 1T   | 83.2       | 54.0* | 80.6 | 92.5         |
| DeepSeek-V4-Flash       | 284B | 73.2       | 45.1  | —    | 90.6         |
| DeepSeek-V4-Pro         | 1.6T | 83.4       | 48.2  | —    | 88.7         |
| MiroThinker-1.7         | 235B | 74.0       | 42.9  | 82.7 | 72.1         |
| Iris-pro                | 397B | 88.6       | 56.4  | —    | 92.9         |
| AREX-Base               | 122B | 82.5       | 52.4  | 85.4 | 89.9         |
| SMALL MODELS (≤40B)     |      |            |       |      |              |
| Tongyi-DeepResearch-30B | 30B  | 43.4       | 32.9  | 70.9 | —            |
| Qwen3.5-35B             | 35B  | 61.0       | 47.4  | 80.0 | 68.5         |
| Quest-35B               | 35B  | 64.6       | 37.2  | 80.8 | —            |
| Apodex-1.0-mini         | 35B  | 71.5       | 46.8  | —    | 82.2         |
| Nex-N2-mini             | 35B  | 74.1       | 37.1  | 82.5 | 87.2         |
| Agents-A1               | 35B  | 75.5       | 47.6  | 96.0 | —            |
| MiroThinker-1.7-mini    | 30B  | 67.9       | 36.4  | 80.3 | 67.9         |
| Iris-mini               | 35B  | 82.2       | 52.3  | —    | 86.9         |
| AREX-Turbo              | 4B   | 70.7       | 40.6  | 81.6 | 78.5         |
| AREX 2.0                | 27B  | 84.0       | 52.6  | 92.2 | 93.8         |

**HLE comparability.** \* Results on the full HLE set. Unmarked results use the text-only subset; the two variants are not directly comparable. A dash indicates an unreported result. All values reproduce the accompanying manuscript's comparison table.